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RECEIVED 05 March 2026

REVISED 03 April 2026

ACCEPTED 15 April 2026

PUBLISHED 20 May 2026

CITATION

Shu J, Zhao C, Chu B, Wang Y, Wu G
and Li B (2026) A closed-loop
framework for PV dynamic hosting
capacity: integrating physics-informed
forecasting and stability constraints.
Front. Energy Res. 14:1823435.
doi: 10.3389/fenrg.2026.1823435

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A closed-loop framework for PV dynamic hosting capacity: integrating physics-informed forecasting and stability constraints

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The escalating penetration of distributed photovoltaic (PV) generation exacerbates the conflict between capacity planning and operational safety, primarily due to source-side stochasticity and grid-side fragility. As a pivotal quantitative benchmark for grid planning and operation, hosting capacity determines the maximum permissible PV integration level that safeguards system reliability against operational violations. However, traditional hosting capacity assessments, often relying on static snapshots, are inadequate for capturing the dynamic coupling among source-side volatility, Static Voltage Stability Margin (SVSM), and active control capabilities. To address these challenges, this paper proposes a closed-loop analytical framework for Dynamic Hosting Capacity (DHC) assessment. First, a physics-informed deterministic forecasting model is developed to accurately track PV output volatility and generate Key Stress Scenarios (KSS) that characterize realistic operational risks. Subsequently, a stability-constrained optimization model is formulated, which explicitly incorporates SVSM as a hard constraint to prevent voltage collapse during critical heavy-load transitions. To further enhance grid accommodation capability, a hierarchical active support mechanism is introduced, employing a centralized-distributed cooperative control strategy to expand the feasible solution space. Case studies on a modified IEEE 33-bus system demonstrate that while SVSM constraints reveal latent stability risks by limiting capacity during peak load periods, the proposed active support effectively relaxes binding constraints, increasing the DHC by approximately 34% compared to passive scenarios. This study provides a rigorous, data-driven methodology for balancing safety margins with asset utilization in active distribution networks.

KEYWORDS

active distribution network, cooperative control, distributed photovoltaic, dynamic hosting capacity, physics-informed deterministic forecasting, static voltage stability

1 Introduction

The global imperative for carbon neutrality is driving a fundamental transformation in power distribution networks, characterized by the rapid integration of Distributed Photovoltaic (DPV) generation (Astero, 2023). While DPV is pivotal for decarbonization, its high penetration introduces substantial stochasticity and volatility to the grid, challenging

the traditional paradigm of “passive distribution network” operation (Zhang et al., 2023). As DPV integration approaches saturation in many regions, the distribution network faces a dual challenge: the “source-side randomness” stemming from intermittent solar irradiance, and the “grid-side stability” constraints caused by bidirectional power flows and reduced system inertia (Gu et al., 2024). In this context, accurately evaluating the Hosting Capacity (HC)—defined as the maximum amount of DPV that can be accommodated without violating operational constraints—has become a critical task for grid planning and operation.

Despite the structural importance of HC assessment, existing methodologies face critical limitations in addressing the fragmentation between forecasting, assessment, and control. Traditional HC assessment often relies on “static snapshot” methods or deterministic time-series analysis based on historical data (Meng et al., 2024). These approaches typically assume that future operating conditions will mirror historical worst-case scenarios (Shen et al., 2025). However, in the volatile context of modern distribution networks—subject to unpredictable weather variations and complex load behaviors—this assumption is frequently violated (Tang et al., 2024). Neglecting the high-frequency volatility and intermittency of DPV output leads to assessment results that are either risky (underestimating uncertainty) or overly conservative (ignoring the probability of extreme events), resulting in the misallocation of grid assets (Li et al., 2023).

Furthermore, a critical gap emerges in the academic discourse regarding the integration of stability constraints and active control into the assessment framework. Conventional assessments predominantly focus on steady-state power quality indicators, such as voltage deviation and thermal loading (Wang et al., 2020; Asaad et al., 2023). However, with the increasing integration of power electronic interfaces, the Static Voltage Stability Margin (SVSM)—the distance to the voltage collapse point—has become a limiting factor, especially during heavy-load, low-irradiance periods (Werkie et al., 2025; Rahman et al., 2021). Recently, data-driven approaches using machine learning have been actively explored to overcome the computational bottlenecks of traditional SVSM analysis, enabling rapid, quasi-real-time stability evaluations (Bogod et al., 2024; Li et al., 2022). Despite these methodological advancements, directly integrating data-driven SVSM as a dynamic hard constraint within hosting capacity optimization remains largely unaddressed. Neglecting SVSM can lead to “static safety but dynamic instability,” creating hidden risks that conventional static or low-resolution methods fail to detect (Yang et al., 2022). Simultaneously, advanced voltage control strategies, such as volt-var control of smart inverters and soft open points, are often treated as operational tools rather than planning resources (Sun

and Qiu, 2021). Recent state-of-the-art studies have demonstrated the immense potential of distributed coordination—such as multi-agent systems and the Alternating Direction Method of Multipliers (ADMM)—in mitigating voltage issues and managing congestion in active distribution networks (ADNs) (Yu et al., 2024). While these pivotal works theoretically prove that active distributed management can expand flexible or dynamic hosting capacity, this “Control-Assessment Decoupling” means that the potential of active support mechanisms to relax binding constraints is rarely quantified in the planning stage (Al-Amin et al., 2026), leading to a significant underestimation of the grid’s true hosting capability (Han, 2025; Trinh and Chung, 2024).

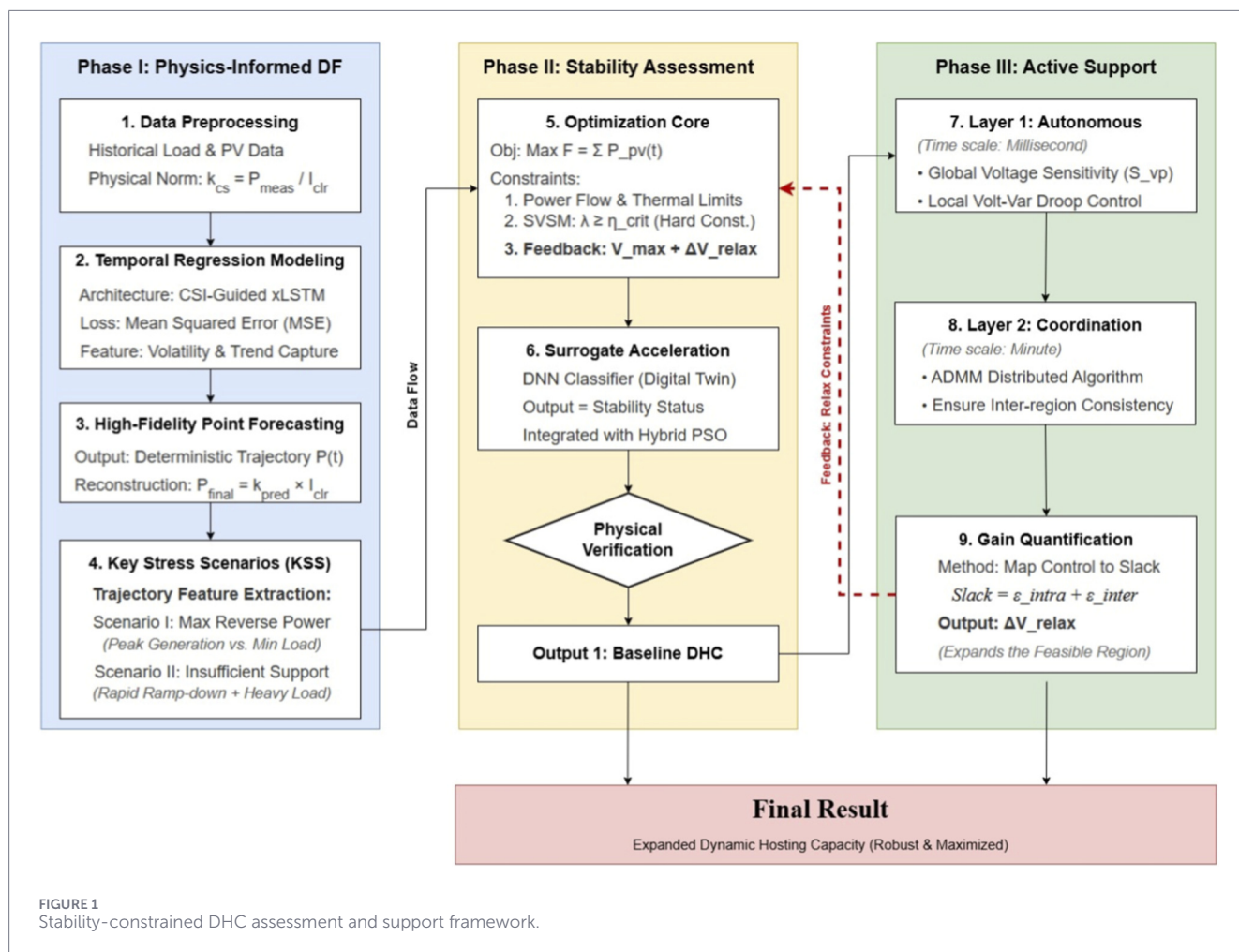
Specifically, the misalignment between operational capabilities and planning assumptions constitutes a fundamental bottleneck. In traditional planning paradigms, voltage regulation devices are often modeled with conservative, fixed setpoints to simplify calculations (Hu et al., 2020). However, in modern ADNs, smart inverters and SVGs possess sub-second response capability that can dynamically suppress voltage excursions (Hamed et al., 2023). Although recent research has made strides in data-driven real-time voltage control (Liu et al., 2025), ignoring this regulation reserve in capacity evaluation essentially renders the assessed hosting capacity a static, worst-case boundary, rather than a dynamic, technically feasible limit. Therefore, establishing a mathematical linkage that maps these high-speed distributed control dynamics into steady-state planning margins is imperative for unlocking the true potential of grid infrastructure.

To bridge these gaps, this paper proposes a closed-loop analytical framework that integrates physics-informed feature extraction, stability-constrained assessment, and hierarchical active support. Unlike previous studies that focus on optimizing a single algorithm, this work aims to reveal the nonlinear coupling mechanisms among prediction, stability, and control. Specifically, we construct a data flow that transforms high-fidelity predicted trajectories into discrete stress scenarios, enforces SVSM as a hard constraint to ensure robustness, and incorporates a centralized-distributed cooperative control strategy to expand the feasible solution space.

The main contributions of this paper are summarized as follows.

1. **High-Fidelity Capture of “Volatility Effect”:** We establish a mapping from temporal volatility features to capacity boundaries, revealing how higher confidence requirements non-linearly contract the Dynamic Hosting Capacity (DHC). By integrating a physics-informed deterministic forecasting model, we generate KSS that capture true operational risks more accurately than arbitrary worst-case assumptions.
2. **Quantification of the “Truncation Effect” of Stability:** By integrating the SVSM into the assessment model, we identify and quantify the capacity reduction caused by stability risks during heavy-loading transitions. This approach addresses the limitation of conventional methods that overlook the bifurcation point proximity, ensuring system robustness under extreme conditions.
3. **Quantification of the “Gain Effect” of Active Support:** We propose a hierarchical control mechanism and quantify its benefit in terms of capacity expansion. Through a centralized-distributed cooperative control strategy utilizing

Abbreviations: ADMM, Alternating direction method of multipliers; ADN, Active distribution network; CPF, Continuation power flow; CSI, Clear sky irradiance; DG, Distributed generation; DHC, Dynamic hosting capacity; DNN, Deep neural network; DPV, Distributed photovoltaic; HC, Hosting capacity; KSS, Key stress scenarios; LSTM, Long short-term memory; MAPE, Mean absolute percentage error; MPPT, Maximum power point tracking; MSE, Mean squared error; PSO, Particle swarm optimization; PV, Photovoltaic; RMSE, Root mean square error; SVG, Static var generator; SVM, Support vector machine; SVSM, Static voltage stability margin; xLSTM, Extended long short-term memory.



the Alternating Direction Method of Multipliers (ADMM), we demonstrate that active voltage support can significantly recover the capacity lost due to uncertainty and stability constraints, effectively converting control capabilities into additional hosting capacity.

2 Overall framework of stability-constrained DHC assessment

To comprehensively address the coupling mechanisms between PV volatility, voltage stability, and active control, this study proposes a unified closed-loop analytical framework structured into three interconnected phases, as illustrated in Figure 1.

First, a physics-informed feature extraction module is constructed to handle source-side stochasticity. By introducing Clear Sky Irradiance (CSI) for physical normalization and employing a CSI-xLSTM network, this phase generates high-resolution deterministic forecast trajectories, which are further condensed into KSS to represent extreme operational risks. Subsequently, a stability-constrained capacity assessment model transforms these scenarios into a baseline DHC. Unlike static methods, this module innovatively integrates the SVSM as a hard constraint to prevent voltage collapse during heavy-load

transitions, utilizing a Deep Neural Network (DNN) surrogate-assisted optimization to ensure computational efficiency. Finally, a hierarchical active support mechanism is introduced to expand the feasible solution space. Through a centralized-distributed cooperative control strategy based on ADMM, this phase creates a feedback loop that quantifies the relaxation effect of active voltage regulation on binding constraints, thereby recovering the capacity lost to uncertainty and stability limits and yielding the robust, maximized Final DHC.

3 Mathematical modeling and algorithm design

3.1 Physics-informed deterministic forecasting

The power output of PV systems is fundamentally governed by astronomical laws, such as the Earth's rotation and revolution (Chen et al., 2023). Consequently, it exhibits significant periodicity and non-stationarity. Utilizing raw data directly for predictive modeling often makes it difficult to decouple deterministic trends from stochastic fluctuations (Hewamalage et al., 2023). To address

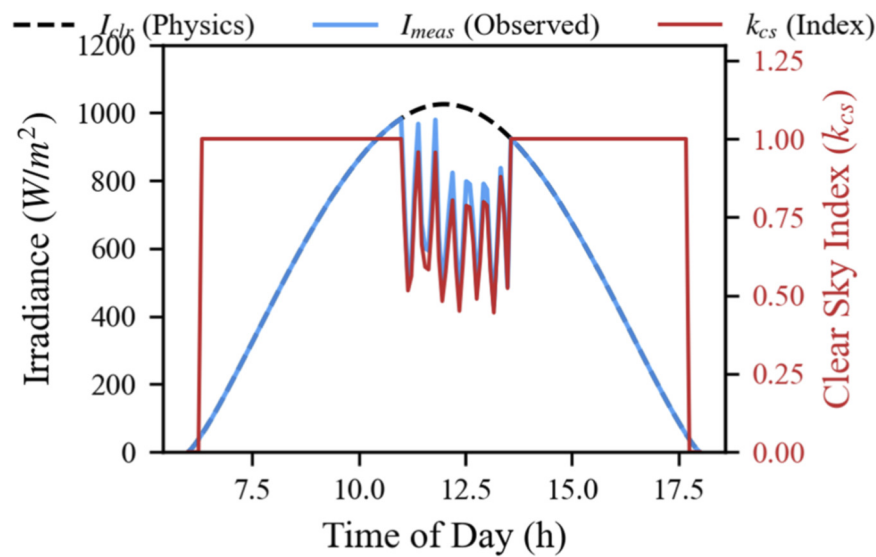


FIGURE 2
Schematic of physics-informed feature extraction.

this, this paper introduces Clear Sky Irradiance (CSI) as a physical benchmark for data denoising.

The clear sky irradiance at time t , denoted as $I_{clr}(t)$, is defined as the theoretical solar radiation intensity under cloudless conditions. Its calculation model is expressed as Equation 1:

$$I_{clr}(t) = I_{sc} \cdot \varepsilon \cdot (\sin \phi \sin \delta + \cos \phi \cos \delta \cos \omega) \quad (1)$$

where I_{sc} represents the solar constant, ε is the correction factor for the sun-earth distance, and ϕ, δ, ω denote the local latitude, solar declination angle, and hour angle, respectively.

Furthermore, the clear sky index $k_{cs}(t)$ is defined to normalize the original measured irradiance $I_{meas}(t)$, as shown in Equation 2:

$$k_{cs}(t) = \frac{I_{meas}(t)}{I_{clr}(t)} \quad (2)$$

As illustrated in Figure 2, the raw irradiance data (I_{meas}) contains both the deterministic diurnal cycle and stochastic fluctuations. By applying the proposed physical normalization, the clear sky index (k_{cs}) effectively removes the bell-shaped trend, isolating the pure stochastic components caused by cloud movement. This stationary feature significantly enhances the learning efficiency of the subsequent neural network.

The transformed sequence eliminates the deterministic components associated with seasonal and diurnal variations. Its numerical fluctuations solely reflect stochastic meteorological processes, such as cloud shading. This transformation significantly enhances data stationarity, thereby facilitating the subsequent model in accurately capturing stochastic uncertainties (Boardman et al., 2025).

Addressing the potential scarcity of historical PV data and the complex temporal dependencies inherent in the series, this study adopts an Extended Long Short-Term Memory (xLSTM) network integrated with physical constraints. The superiority of xLSTM

over standard recurrent architectures lies in its exponential gating mechanism and revised memory structures. Solar irradiance data is characterized by a superposition of low-frequency diurnal trends and high-frequency turbulence caused by cloud motion. Standard LSTMs often struggle to retain high-frequency signal details over long sequence horizons due to the forgetting gate bias. The xLSTM, by contrast, enhances the gradient flow and memory capacity, allowing the model to simultaneously capture the macro-trend of clear-sky cycles and the micro-fluctuations of rapid shading events. This dual-capture capability is crucial for identifying the insufficient voltage support scenarios.

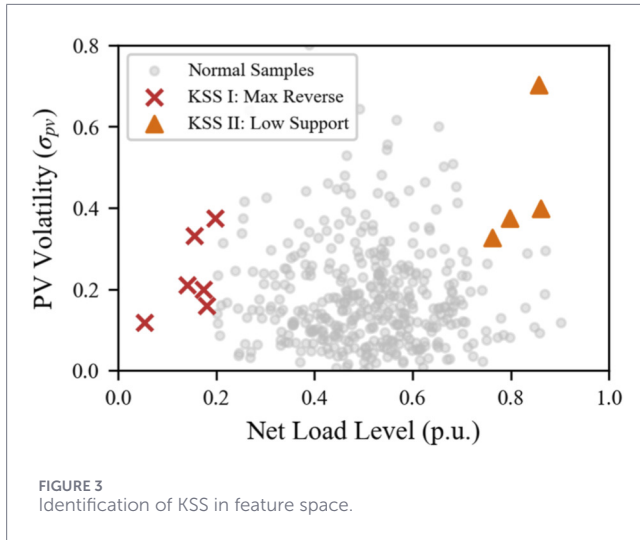
The proposed framework comprises an embedding network, a recovery network, a generator, and a discriminator. The generator utilizes random noise to synthesize latent trajectories, while the discriminator is tasked with distinguishing between real and synthetic trajectories. The incorporation of xLSTM enables the model to capture long-term temporal correlations in PV output more effectively. The min-max game objective function of the model is formulated as Equation 3:

$$\mathcal{L} = \frac{1}{N} \sum_{t=1}^N (P_{real}(t) - P_{pred}(t))^2 \quad (3)$$

The objective function aims to minimize the Mean Squared Error (MSE) between the predicted PV output $P_{pred}(t)$ and the actual measurement $P_{real}(t)$.

Finally, based on the high-precision deterministic forecast, KSS are identified to capture the most restrictive operating conditions. Instead of relying on static worst-case assumptions, we extract critical time steps by analyzing the temporal alignment between the predicted PV curve and the load profile.

1. Maximum Reverse Power Flow Scenario: Defined at the time step t_{max} where the net injected power $P_{net}(t) = P_{pv}^{pred}(t) - P_{load}(t)$ reaches its maximum positive value. This captures the risk of voltage violation due to surplus generation.



2. Insufficient Voltage Support Scenario: Defined at the time step $t_{\min}$ where a rapid drop in PV output coincides with a heavy load level. This characterizes the risk of voltage collapse when active support is suddenly withdrawn.

The distribution of these scenarios is visualized in Figure 3. Unlike traditional methods that rely on a single worst-case snapshot, our approach identifies distinct clusters of operational risks. The Reverse Flow scenarios (marked in red) concentrate in the low-load region, while Ramp Risk scenarios (marked in orange) appear in the high-load, high-volatility quadrant. This selective extraction ensures that the subsequent stability assessment focuses on the most binding boundary conditions.

3.2 Stability-constrained capacity assessment model

Based on the KSS generated in the feature extraction stage, a dynamic optimization model is established to determine the DHC. Unlike traditional assessment methods that predominantly focus on steady-state limits, this framework innovatively integrates the SVSM as a core constraint. The primary objective is to maximize the total permissible penetration of DPV across the network over the evaluation horizon T , formulated as Equation 4:

$$\max F = \sum_{t=1}^T \sum_{i \in \mathcal{N}_{PV}} P_{PV,i,t} \quad (4)$$

where $\mathcal{N}_{PV}$ represents the set of PV nodes, $P_{PV,i,t}$ represents the active power output of the PV unit at node i at time t . This objective is subject to three categories of operational constraints under the identified KSS: (i) Power Flow Constraints, which ensure the balance of active and reactive power according to Kirchhoff's laws; (ii) Voltage Deviation Constraints, requiring nodal voltages $V_{i,t}$ to remain within the permissible range $[V_{\min}, V_{\max}]$; and (iii) Thermal Constraints, ensuring branch power flows do not exceed line ratings.

A critical limitation of conventional assessment is the neglect of voltage stability in high-impedance networks. The operating point may approach the voltage collapse point even when voltage

magnitudes appear normal. To address this, the SVSM constraint is enforced. The SVSM is quantified by the loadability factor λ , defined as the multiplier by which the net load can be increased before the power flow Jacobian matrix becomes singular. The constraint is expressed as Equation 5:

$$\lambda_{\max,t} \geq \eta_{\text{threshold}}, \forall t \in T \quad (5)$$

where $\lambda_{\max,t}$ is the maximum loading margin calculated via Continuation Power Flow (CPF), and $\eta_{\text{threshold}}$ is the required safety margin.

Solving this problem is computationally intensive due to the iterative nature of CPF. To achieve quasi-real-time assessment, a surrogate-assisted strategy is employed. A DNN is trained offline to map the input vector $\mathbf{x} = [P_{PV}, P_{\text{load}}]$ to the binary stability status S . The optimization search is then accelerated by a Hybrid Particle Swarm Optimization (PSO) algorithm, where the fitness function incorporates the surrogate prediction, as shown in Equation 6:

$$\text{Fitness}(\mathbf{x}) = \begin{cases} \sum P_{PV}, & \text{if } f_{DNN}(\mathbf{x}) \geq \tau \\ -\infty, & \text{otherwise} \end{cases} \quad (6)$$

where f_{DNN} is the probability output of the surrogate model and τ is the decision threshold (set to 0.5 to align with the perfectly balanced training dataset). To strictly prevent false negatives and ensure a conservative assessment, the surrogate acts as a rapid pre-filter during the heuristic search. Subsequently, any optimal candidate passing this threshold is subjected to the exact physical CPF model for final verification (as shown in Figure 1), thereby balancing computational speed with absolute operational accuracy.

Specifically, the DNN surrogate employs a MLP architecture comprising three hidden layers with 128, 64, and 32 neurons, respectively. The ReLU activation function is applied across all hidden layers to facilitate gradient flow, while a Sigmoid activation at the output layer generates the binary classification probability. To train this surrogate, an offline dataset comprising 50,000 samples was generated utilizing Latin Hypercube Sampling (LHS) to explore the comprehensive operational space of P_{PV} and P_{load} . The rigorous CPF algorithm was executed for each sample to assign the ground-truth stability label. To prevent model bias caused by the natural scarcity of unstable operating points, the dataset was balanced using a hybrid approach: conducting high-density targeted sampling near the theoretical stress boundaries, followed by the application of the Synthetic Minority Over-sampling Technique (SMOTE). This ensured an exact 1:1 class distribution between stable and unstable scenarios in the final training corpus.

To validate the fidelity of the surrogate model, Figure 4 visualizes its decision boundary in a two-dimensional projection. The model successfully demarcates the Safe Zone from the Risk Zone, correctly identifying that system instability is triggered by the combination of high system load and insufficient reactive power support. This high-precision classification capability justifies its use in the online optimization loop.

3.3 Hierarchical active support and capacity feedback mechanism

To convert operational voltage regulation potential into assessable hosting capacity, this paper establishes a hierarchical

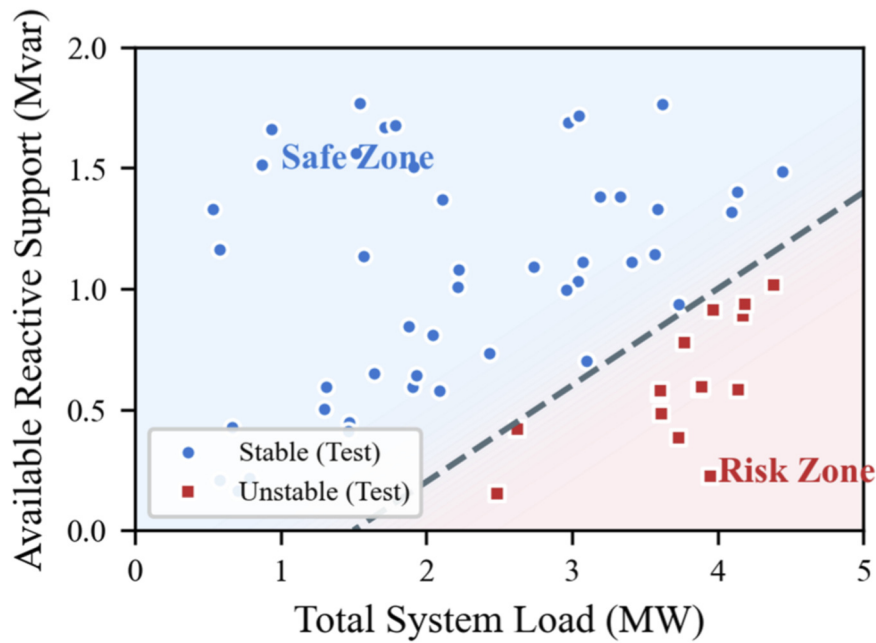


FIGURE 4
Visualization of the stability decision boundary learned by the surrogate model.

active support mechanism. Unlike traditional passive assessments, this framework actively utilizes controllable resources to expand the feasible solution space. The mechanism consists of a two-stage cooperative control strategy and a quantification module that links control performance to capacity planning.

3.3.1 Two-stage hierarchical cooperative control

To balance response speed with global optimality, a distributed control model is formulated. The primary objective is to minimize system voltage deviation and power losses, subject to power flow and device operational constraints, as shown in Equations 7–13:

$$\begin{aligned} \min F &= \sum (S_k^V)^2 + \sum \gamma_i P_i^{\text{loss}} \\ \text{s.t. } V_i^2 &= g_i(P, Q) \\ (P, Q) &\in \Omega \end{aligned} \quad (7)$$

$$\underline{P}_i^{\text{DG}} \leq P_{i,t}^{\text{DG}} \leq \bar{P}_i^{\text{DG}}, Q_i^{\text{DG}} \leq Q_{i,t}^{\text{DG}} \leq \bar{Q}_i^{\text{DG}} \quad (8)$$

$$\underline{P}_i^{\text{PV}} \leq P_{i,t}^{\text{PV}} \leq \bar{P}_i^{\text{PV}}, (P_{i,t}^{\text{PV}})^2 + (Q_{i,t}^{\text{PV}})^2 \leq (S_i^{\text{PV}})^2 \quad (9)$$

$$Q_i^{\text{SVG}} \leq Q_{i,t}^{\text{SVG}} \leq \bar{Q}_i^{\text{SVG}} \quad (10)$$

$$\underline{V}_{i,t} \leq V_{i,t} \leq \bar{V}_{i,t} \quad (11)$$

$$P_{i,t} = P_{i,t}^{\text{Load}} - P_{i,t}^{\text{PV}} - P_{i,t}^{\text{DG}} \quad (12)$$

$$Q_{i,t} = Q_{i,t}^{\text{Load}} - Q_{i,t}^{\text{SVG}} - Q_{i,t}^{\text{PV}} - Q_{i,t}^{\text{DG}} \quad (13)$$

Solving this optimization problem centrally is computationally prohibitive for real-time applications. Therefore, a hierarchical

strategy combining millisecond-level autonomy and minute-level coordination is adopted.

3.3.1.1 Layer 1: millisecond-level autonomous response

To cope with rapid solar irradiance fluctuations, a fast response mechanism based on global voltage sensitivity is employed. The sensitivity matrix S , derived from the inverse Jacobian of the polar coordinate power flow equations, quantifies the direct response of nodal voltage to power injections:

$$\begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix} = J^{-1} \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} S^{\theta\theta} & S^{\theta V} \\ S^{V\theta} & S^{VV} \end{bmatrix} \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} \quad (14)$$

Based on the system Lagrangian of Equation 10, the gradient of the objective function with respect to active and reactive power injections serve as the control signal. The global sensitivities $S_{i,t}^P$ and $S_{i,t}^Q$ are derived as:

$$S_{i,t}^P = \frac{\partial L}{\partial P_{i,t}^{\text{DG}}} = S_{i,t}^{P_{\text{loss}}} + \gamma_i P_i^{\text{loss}} \quad (15)$$

$$S_{i,t}^Q = \frac{\partial L}{\partial Q_{i,t}^{\text{DG}}} = S_{i,t}^{Q_{\text{loss}}} + \gamma_i Q_i^{\text{loss}} \quad (16)$$

In the first stage, these sensitivity indices allow local devices to autonomously adjust their output in the direction of steepest descent for voltage violation mitigation, achieving quasi-instantaneous regulation without waiting for global optimization convergence.

However, it is crucial to acknowledge that the voltage sensitivity matrices defined in Equations 14–16 inherently rely on a linear approximation of the power flow equations. While highly effective for small-magnitude, high-frequency stochastic fluctuations, this linear assumption inevitably degrades as the system operating point approaches the stability boundary characterized

TABLE 1 Configuration of the modified IEEE 33-bus system.

Device type	Node index (bus no.)	Capacity/Parameters	Control mode
DPV 1	13	0.3 MW	MPPT/Droop
DPV 2	17	0.2 MW	MPPT/Droop
DPV 3	24	0.4 MW	MPPT/Droop
DPV 4	30	0.2 MW	MPPT/Droop
DPV 5	32	0.1 MW	MPPT/Droop
SVG 1	17	±0.8 Mvar	Constant Q/Remote
SVG 2	32	±0.4 Mvar	Constant Q/Remote
Load model	All load buses	Peak: 3.715MW + j2.3 Mvar	Constant power (P,Q)

by strong nonlinearities—such as those encountered during the KSS identified in Section 3.1. However, within the proposed hierarchical framework, this sensitivity-based approximation serves exclusively as the Layer one response (millisecond-level), prioritizing computational speed to track rapid solar transients. The inherent “approximation drift” caused by intense nonlinearity under heavy-load stress is systematically corrected by Layer 2. The subsequent minute-level ADMM coordination is explicitly formulated as a non-linear optimization model (Equation 7), which recalculates the exact global power flow and eliminates residual voltage violations caused by local greedy linear control, thereby guaranteeing operational safety even at the stability margins.

3.3.1.2 Layer 2: minute-level ADMM coordination

While local autonomy handles high-frequency fluctuations, it may lead to suboptimal solutions or boundary violations between control areas. To ensure global consistency, the ADMM is applied.

By introducing boundary variables (x_{bound}) and consistency variables (z), the coupling constraints between regions are decoupled. The augmented Lagrangian function for each region k is constructed as Equation 17:

$$L(x_k, z, \lambda) = f(x_k) + \sum_{k \in N_k} \left(\lambda_{kl}^T (x_{k,bound} - z_{kl}) + \frac{\rho}{2} \|x_{k,bound} - z_{kl}\|_2^2 \right) \quad (17)$$

Through the iterative update of primal variables, dual multipliers (λ), and consistency variables, the algorithm eliminates residual voltage violations caused by local greedy control, ensuring the system operates within safety margins.

3.3.2 Quantifying control gain for DHC assessment

This section connects operational control with planning and evaluation. Unlike the traditional DHC evaluation model before, which treats voltage as a rigid constraint, the multi-region, multi-time-scale collaborative control proposed in this section provides

active voltage support capabilities for the distribution network. To quantify the improvement in PV hosting capacity resulting from this capability, an active support evaluation model considering voltage relaxation constraints is established. Its core principle is to transform the adjustment potential of the control system, operating autonomously at the second level and collaboratively at the minute level, into dynamic relaxation variables for voltage constraints in the evaluation model, thereby mathematically expanding the feasible region.

3.3.2.1 Quantifying control capability as evaluation slack

First, for the layer-internal, second-level autonomous method based on global sensitivity, the ultimate capability of this dynamic process can be statically quantified in the evaluation model. The voltage regulation capability provided by layer-internal autonomy for node i can be expressed as the weighted aggregation of the remaining reactive power capacity of all controllable regulation resources within region k with respect to the voltage sensitivity of that node, i.e., as shown in Equation 18:

$$\varepsilon_{i,intra} = \sum_{j \in \Omega_{ctrl}} (S_{ij}^{PV} \cdot \Delta P_{j,t} + S_{ij}^{QV} \cdot \Delta Q_{j,t}) \quad (18)$$

where Ω_{ctrl} represents the set of controllable devices belonging to the same control area as node i ; S_{ij}^{PV} , S_{ij}^{QV} represent the active power-voltage sensitivity and reactive power-voltage sensitivity elements calculated; and $\Delta P_{j,t}$, $\Delta Q_{j,t}$ represents the maximum active power and reactive power adjustment amounts that can be utilized at node j .

3.3.2.2 Interlayer collaborative relaxation variable quantization

For multi-region minute-level coordination based on ADMM, the additional voltage improvement capability $\varepsilon_{i,inter}$ brought about by inter-layer collaboration is obtained by solving a simplified linear programming subproblem aimed at maximizing regional boundary voltage support. Specifically, to obtain the inter-layer relaxation component $\varepsilon_{i,intra}$ for target node i , we construct the following linear programming subproblem aimed at maximizing boundary voltage support. This model simulates the ultimate effect of the ADMM algorithm in extreme operating conditions, where network resources are shifted towards heavily loaded regions, as shown in Equations 19–21:

$$\max \varepsilon_{i,intra} = \sum_{b \in B_K} (S_{ib}^{QV} \cdot \Delta Q_b + S_{ib}^{PV} \cdot \Delta P_b) \quad (19)$$

$$s.t. 0 \leq \Delta Q_b \leq \Delta Q_{b,N}^{\max} \quad (20)$$

$$V_b^{\min} \leq V_b^{ner} - S_{bb}^N \Delta Q_b \leq V_b^{\max} \quad (21)$$

where B_K represents the set of all boundary nodes in region K ; Equation 20 represents the domain support capacity constraint, meaning that the boundary injected power ΔQ_b cannot exceed the remaining capacity $\Delta Q_{b,N}^{\max}$ of the adjacent nodes; and Equation 21 represents the ADMM collaborative consistency relaxation constraint, meaning that the boundary voltage change due to injected power cannot cause the domain voltage V_b^{ner} to exceed its limits.

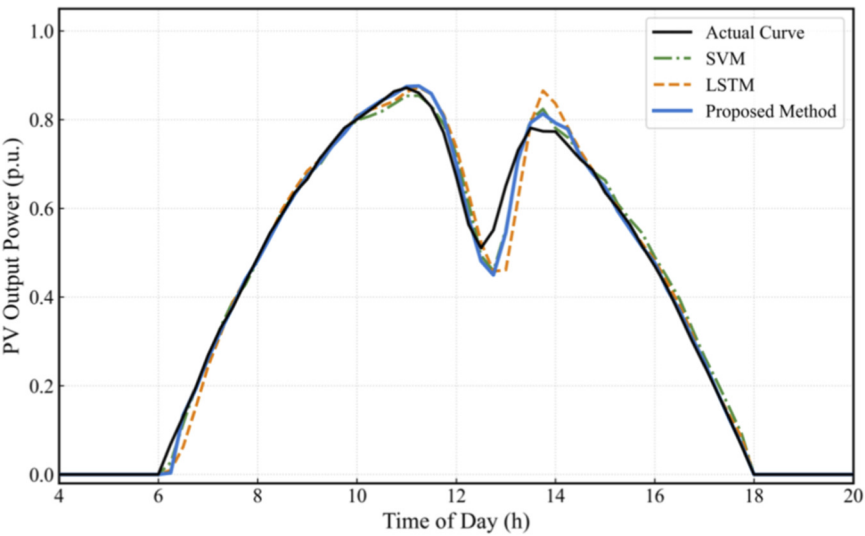


FIGURE 5
Comparison of PV power forecasting results among different methods.

TABLE 2 Performance comparison of feature extraction methods.

Model	RMSE	MAE	MAPE (%)
SVM	0.029	0.021	6.34
LSTM	0.045	0.027	8.06
Proposed method	0.027	0.015	4.53

4 Experiment results

To validate the effectiveness of the proposed framework, numerical simulations were performed on a modified IEEE 33-bus distribution system. The dataset covers a 60-day operating period with a 15-min resolution (96 steps/day), incorporating multi-dimensional features such as solar irradiance, temperature, and wind speed. This period corresponds to the summer months (July–August), deliberately selected to encapsulate the most restrictive operational conditions. During this season, the grid confronts overlapping stress factors: annual peak load demands driven by cooling requirements, coupled with high-amplitude PV volatility caused by frequent summer convective cloud shading. A CSI-xLSTM generative model is trained to capture the stochastic volatility of PV generation. The system configuration, including the spatial distribution of DG and SVGs, is detailed in Table 1. All load buses are modeled as constant power (P, Q) nodes to reflect realistic demand profiles.

The predictive performance of the proposed method is compared against traditional baselines, including Support Vector Machine (SVM) and Long Short-Term Memory (LSTM) networks. As illustrated in Figure 5, the baseline models exhibit noticeable phase lags and smoothing effects during rapid irradiance transitions. In contrast, the CSI-xLSTM framework tracks the actual trajectory with high fidelity, effectively capturing the sharp power dip around 12:30 caused by cloud shading.

Table 2 provides a comprehensive quantitative validation of the model’s performance. The proposed CSI-xLSTM framework demonstrates superior predictive fidelity, minimizing the Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) to 0.027% and 4.53%, respectively. This substantial reduction in prediction error creates a robust foundation for the subsequent assessment, ensuring that the generated KSS capture genuine stochastic volatility rather than numerical artifacts.

Based on the generated KSS, the DHC is evaluated. Figure 6 contrasts the results between the traditional method (steady-state voltage constraints only) and the proposed stability-constrained framework.

Distinctly, as the system approaches peak load (approx. 19:00), the traditional method erroneously suggests a high hosting capability. However, the proposed framework identifies a severe truncation effect on capacity peaks. This occurs because the heavy load profile drives the operating point toward the “nose” of the P-V curve, activating the SVSM hard constraint. The hatched Risk Zone in Figure 6 represents capacity that is statically feasible but dynamically unstable, highlighting the necessity of stability constraints.

From a physical perspective, it is imperative to note that the severity of the “Truncation Effect” observed in Figure 6 is theoretically sensitive to the assumed load characteristics. In this study, as detailed in Table 1, loads are modeled as constant power (PQ) nodes. This assumption acts as the most conservative scenario for voltage stability; under a pure PQ model, an incipient voltage drop forces the load to draw higher current to maintain constant power, which subsequently triggers heavier line drops and accelerates the approach to the P-V curve “nose” point. Conversely, if voltage-dependent loads—such as the constant impedance components typical in ZIP models—were prevalent, the power demand would naturally diminish alongside nodal voltage sags. This inherent self-relieving property would delay voltage collapse and increase the physical SVSM. Consequently, the presence of voltage-dependent loads would render the stability constraint less binding, theoretically mitigating the truncation effect and further

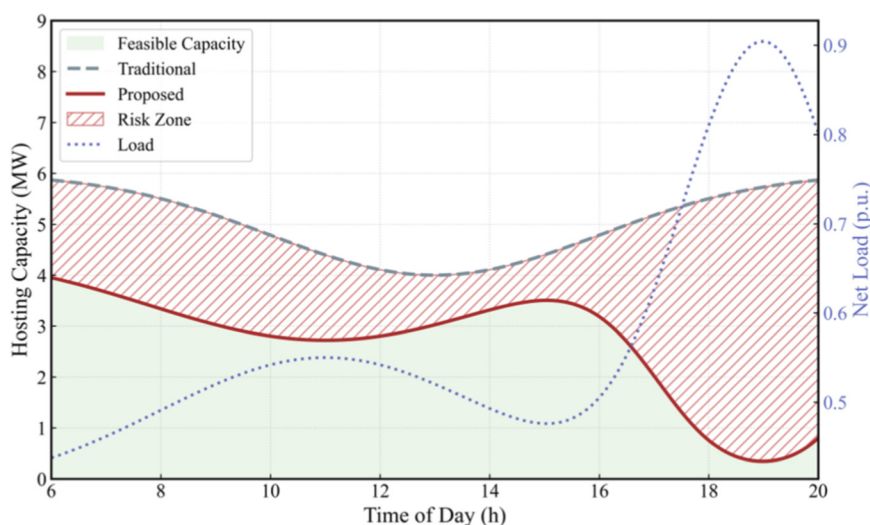


FIGURE 6
Impact of SVSM constraints on dynamic hosting capacity and the identification of hidden stability risks.

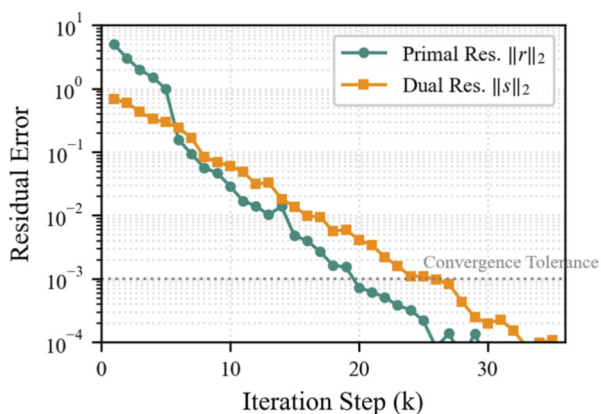


FIGURE 7
Convergence performance of the distributed ADMM coordination algorithm.

expanding the DHC. By adopting the strict PQ model, the proposed framework deliberately evaluates the conservative lower bound of hosting capacity to guarantee robust planning margins, though future operational models may integrate dynamic ZIP characteristics for refined assessments.

Before evaluating the final hosting capacity, the computational performance of the hierarchical coordination layer is verified. As shown in Figure 7, the ADMM-based algorithm exhibits robust convergence on the modified IEEE 33-bus system. Both primal and dual residuals decrease to the tolerance level within 30 iterations, confirming that the proposed active support mechanism can be executed efficiently within the minute-level operational window.

To achieve this consistent convergence across highly variable operational conditions, the penalty parameter ρ was not fixed empirically but implemented with a dynamic residual-balancing

strategy. Specifically, ρ is adaptively updated at the end of each iteration to maintain the primal and dual residuals within a similar magnitude. If the primal residual significantly exceeds the dual residual, ρ is increased to penalize boundary consistency violations more heavily; conversely, it is decreased if the dual residual dominates. This self-tuning mechanism eliminates the need for manual parameter tuning and successfully prevents algorithmic stalling, facilitating the rapid convergence observed.

To verify the effectiveness of the hierarchical collaborative control strategy proposed in this study for increasing the available capacity of distributed photovoltaic systems in distribution networks, this section sets up two sets of comparative experiments. By simulating photovoltaic DHC under different modes in the IEEE 33-bus system, the expansion effect of the active grid support mechanism on the feasible region boundary is quantitatively analyzed.

This experiment selects typical daily load and photovoltaic output curves, and compares the impact of the following two operating modes on the DHC evaluation results under the same prediction uncertainty level (confidence level $\alpha = 90\%$) and SVSM constraint ($\text{SVSM} \geq 10\%$).

To evaluate the spatial impact of the proposed control framework, Figure 8 illustrates the nodal voltage profiles along the IEEE 33-bus feeder during a critical period of high PV generation. Under the Passive Integration (Mode 1), the system suffers from severe over-voltage violations at two distinct clusters—Buses 10–17 and 31–32—with peak magnitudes reaching approximately 1.08 p. u., significantly exceeding the statutory 1.05 p. u. safety limit. These spatial violations constitute the primary bottleneck that truncates the hosting capacity in traditional assessments.

In contrast, the Proposed Active Support (Mode 2)—which integrates the hierarchical coordination mechanism—effectively suppresses these voltage excursions. As shown in the blue profile, the nodal voltages across the entire feeder are regulated back within the permissible 1.05 p. u. boundary. The shaded “Voltage Violation Mitigation” area quantifies the effectiveness

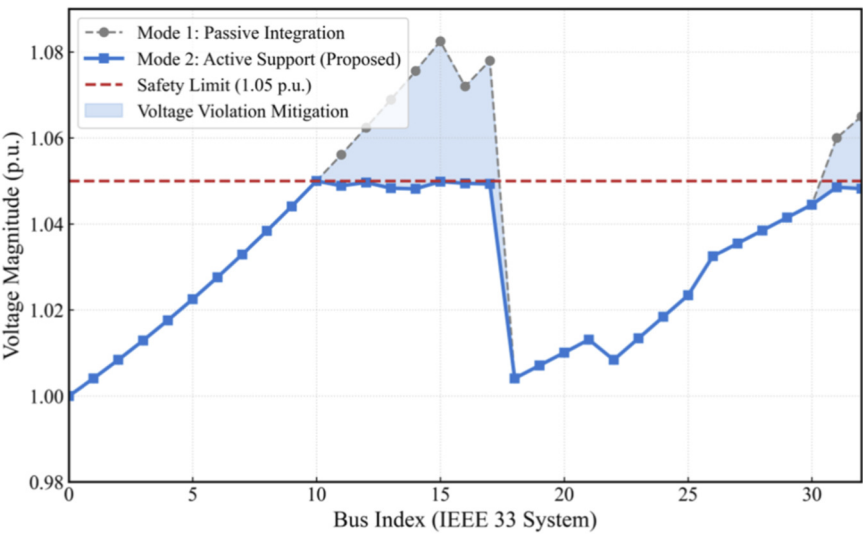


FIGURE 8 Comparison of voltage profiles along the IEEE 33-bus system under passive and active support modes.

TABLE 3 Sensitivity of dynamic hosting capacity to different SVSM safety margins.

Stability margin η	Static DHC (MW)	Stability-constrained DHC (MW)	Capacity reduction (%)	Min. Nodal voltage (p.u.)
$\eta = 0.05$	6.42	5.85	8.8	0.962
$\eta = 0.10$	6.42	5.12	20.2	0.968
$\eta = 0.15$	6.42	4.45	30.6	0.975
$\eta = 0.20$	6.42	3.82	40.5	0.981

TABLE 4 Quantitative enhancement of hosting capacity under different modes.

Control strategy (mode)	Total daily energy (MWh)	Max. Instantaneous HC (MW)	Avg. Loadability factor (λ)	Improvement rate (%)
M1: Passive integration	18.5	2.45	10.05	-
M2: Local droop control	21.2	3.10	11.12	+14.6%
M3: Cooperative (proposed)	24.8	4.25	12.28	+34.1%

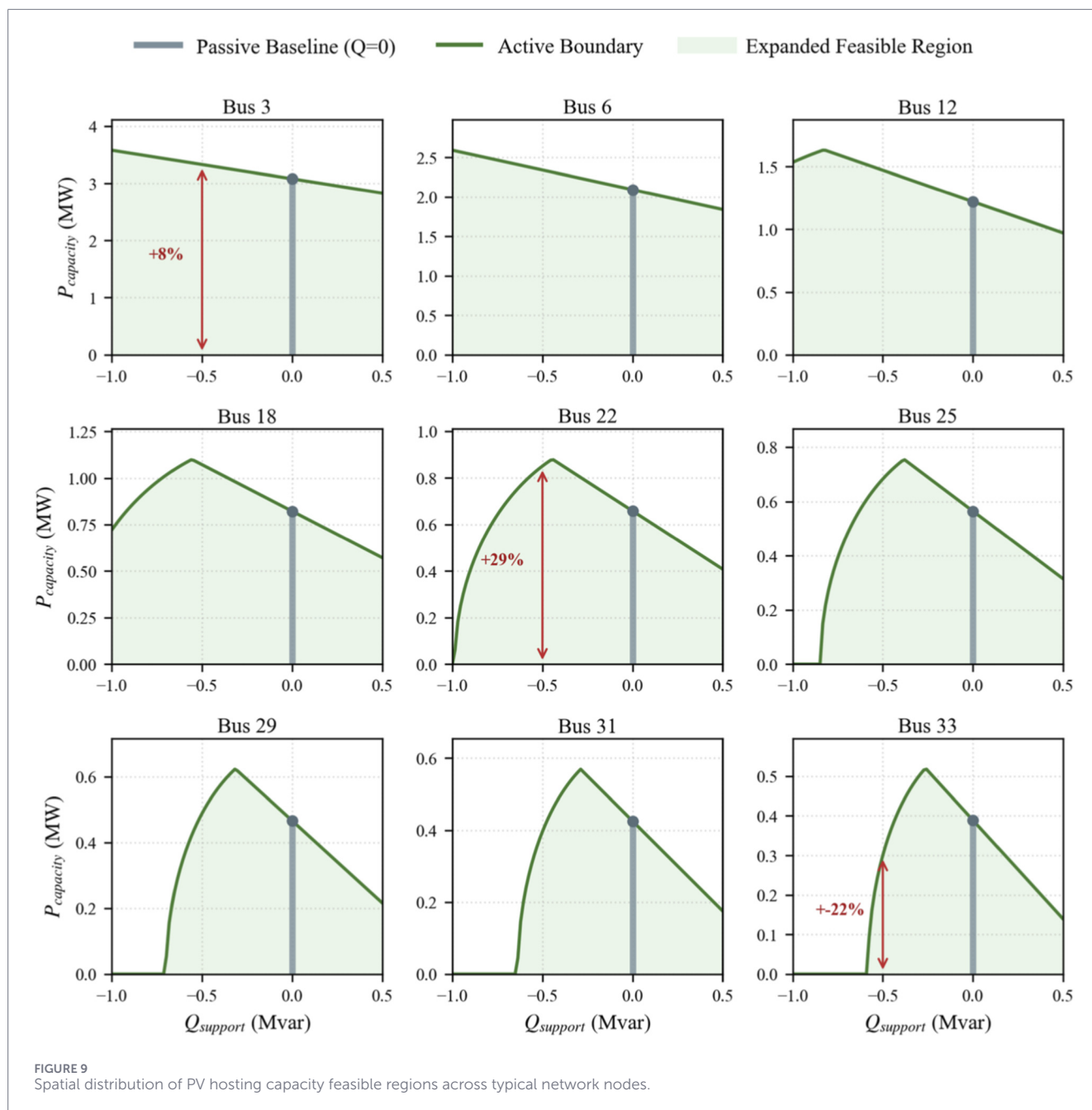
of the proposed cooperative control in relaxing binding voltage constraints. This spatial regulation capability is the fundamental driver for the “Gain Effect” observed in the capacity expansion results, successfully converting control potential into additional PV integration headroom.

As quantified in Table 3, there is a clear trade-off between grid stability and PV accommodation. While traditional assessments remain constant, the proposed stability-constrained DHC non-linearly contracts as the required safety margin η increases. Notably, even when nodal voltages are within the 0.95–1.05 p. u. range, a high stability requirement ($\eta = 0.20$) can reduce the permissible capacity by over 40%, highlighting the necessity of including SVSM in planning.

In the passive scenario (M1), significant voltage violations occur at buses 10–17 and 31–32, with peak magnitudes exceeding 1.08

p. u., which directly limits the further integration of DPV. By contrast, the proposed cooperative control effectively regulates the nodal voltages back within the statutory 1.05 p. u. limit. The shaded “Voltage Violation Mitigation” area highlights the effectiveness of the centralized-distributed coordination in relaxing binding voltage constraints.

Table 4 provides a quantitative summary of the hosting capacity enhancement under three different modes. To ensure reproducibility, the local droop control baseline (M2) is configured in strict accordance with the IEEE 1547–2018 standard for smart inverter Volt-Var functions, featuring a voltage deadband of ± 0.02 p. u. and triggering maximum reactive power support at the 0.95 and 1.05 p. u. statutory limits. While M2 provides a moderate improvement of 14.6%, the proposed hierarchical cooperative control (M3) achieves a substantial



gain of 34.1%, increasing the total daily energy integration to 24.8 MWh. Furthermore, M3 maintains the highest average loadability factor (λ), demonstrating that active control not only expands the feasible capacity region but also enhances the overall grid stability. These results validate the “Gain Effect” of active support, proving that coordinating local autonomous responses with global optimization can significantly recover the hosting capacity previously lost to uncertainty and stability risks.

To further reveal the spatial universality of the proposed method, Figure 9 illustrates the P-Q operating envelopes for nine representative nodes across the network. While nodes near the substation show limited marginal gains due to strong grid stiffness, the weak nodes at the feeder ends exhibit a dramatic expansion in

their feasible regions. By actively utilizing the reactive power support (negative X-axis), the proposed framework effectively overcomes the voltage bottlenecks that previously constrained these nodes, verifying the Gain Effect reported in Table 4.

5 Discussion

5.1 The necessity of coupling prediction and stability

The results underscore the critical nonlinear coupling among prediction uncertainty, voltage stability, and active control. Traditional assessments often treat these elements in isolation,

leading to capacity estimates that are either overly optimistic (ignoring stability) or overly conservative (ignoring control). Our comparative analysis reveals that during peak load transitions, the system's binding constraint shifts from simple voltage deviation to voltage stability margins. The proposed CSI-xLSTM framework outperforms traditional baselines by effectively capturing high-frequency fluctuations, thereby generating realistic stress scenarios that trigger these stability limits. This proves that high-fidelity forecasting is not merely a data requirement but a prerequisite for identifying hidden stability risks.

From a practical deployment standpoint, determining the optimal safety margin (η) explored in Table 3 should not be a one-size-fits-all static assumption. Distribution System Operators (DSOs) are advised to determine this threshold through a localized risk-cost analysis. Practically, the optimal η should be dynamically tailored based on the empirical error bounds of the regional forecasting tools—where higher prediction uncertainty necessitates a larger stability buffer—and the economic trade-off between the financial cost of limiting PV integration and the reliability penalties of a potential voltage collapse.

5.2 The economic value of Virtual Capacity

The Gain Effect observed in the case studies—a 34.1% increase in hosting capacity—translates directly into economic benefits. By utilizing the proposed hierarchical control to relax binding constraints, utilities can defer capital-intensive grid reinforcements. The derived “Voltage Violation Mitigation” area can be interpreted as Virtual Capacity created by software algorithms rather than hardware investment. This suggests that future distribution network planning should prioritize the deployment of communication and control layers alongside physical infrastructure.

5.3 Scalability and future work

While the current study validates the framework on a balanced IEEE 33-bus system, its application to larger, unbalanced three-phase networks presents additional challenges. The ADMM-based coordination demonstrates good scalability due to its distributed nature, but communication latency in real-world deployments could degrade the millisecond-level response assumed in Layer 1. Future work will focus on integrating robust control strategies that account for packet loss and delays. Additionally, expanding the SVSM constraints to account for phase imbalance will be crucial for practical implementation in low-voltage residential feeders.

6 Conclusion

This study proposes a closed-loop analytical framework for Dynamic Hosting Capacity assessment that integrates physics-informed feature extraction, stability-constrained optimization, and hierarchical active support. Experimental validation on the IEEE 33-bus system confirms that while SVSM constraints are essential for preventing voltage collapse during critical load transitions, the proposed centralized-distributed cooperative control effectively counteracts the resulting capacity contraction. By recovering

approximately 20%–34% of the hosting capacity compared to passive scenarios, this methodology provides a rigorous, data-driven tool for Distribution System Operators to balance operational safety with asset utilization, facilitating a paradigm shift from passive acceptance to active, stability-aware renewable integration.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

JS: Investigation, Writing – original draft, Conceptualization, Writing – review and editing, Methodology. CZ: Methodology, Writing – original draft, Formal Analysis, Investigation. BC: Methodology, Investigation, Supervision, Writing – original draft, Project administration, Funding acquisition. YW: Validation, Writing – review and editing, Methodology, Data curation, Visualization. GW: Formal Analysis, Writing – original draft, Software, Investigation. BL: Writing – review and editing, Validation, Methodology, Visualization.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This research was funded by Inner Mongolia Power Company, grant number No.: 2025-3-9 and the APC was funded by Inner Mongolia Power Company 2025 Science and Technology Project “Research and Application of Key Technologies for Distributed PV Hosting Capacity Assessment and Voltage Active Support in Distribution Networks for New-Type Power Systems”. The funder was not involved in the study design, collection, analysis, interpretation of data, the writing of this article, or the decision to submit it for publication.

Conflict of interest

Authors JS, CZ, BC, YW, GW, and BL were employed by Xuejiawan Power Supply Company of Inner Mongolia Electric Power (Group) Co. Ltd.

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